CS 135 – Programming Assignment 3

Case File: Mr. Ed's 75th Birthday Investigation

1. Background Story

Mr. Ed, aged 75, was celebrating his birthday at home with five close friends:

1. Mr. Brown – His lawyer
2. Parker – His violin teacher
3. Mark – His college buddy (same age group)
4. Dayan – His 22-year-old chess partner
5. Ms. Elena – His pen friend (meeting in person for the first time)

Suddenly, Mr. Ed is found in the bathroom with a severe head injury. Emergency services are on the way. Until then, you must logically analyze the situation and determine:

- Was this likely an accident?
- Or was it likely murder?
- If murder, who is the strongest suspect?

You will implement this investigation using only conditional logic.

2. Learning Outcomes

By completing this assignment, you will:

- Practice structured decision-making using if, else if, and else
- Implement nested conditional logic correctly
- Convert user inputs into logical evidence variables
- Understand multi-layer reasoning in programming
- Improve indentation, readability, and logical structure
- Avoid "one-line condition dumping" and instead build clean logic blocks

This assignment is about thinking like a programmer, not just writing code.

3. Restrictions:

You may use:

- if
- else if
- else
- nested conditionals
- bool, int, string
- cin and cout

You may **NOT** use:

- loops (for, while, do-while)
- arrays or vectors
- switch
- functions
- ternary operator
- combining everything into one giant condition

4. Program Structure Overview

Your program must be divided into three major parts:

i. PART 1: Scene Analysis (Accident vs Murder)
ii. PART 2: Suspect Evaluation
iii. PART 3: Interview Questions & Evidence Mapping

4.1. PART 1−Scene Analysis (Accident vs Murder)

Before analyzing suspects, evaluate the scene.

Ask the user for the following inputs (1 = Yes, 0 = No) for each of these boolean variables:

- wetFloor
- brokenVase
- heavyObjectMissing
- medicineScattered

4.1.1. <u>Scene Decision Rules</u>

You must determine:

Use nested logic to decide:

- If heavyObjectMissing == 1
  → likelyMurder = true
- Else if brokenVase == 1
  - If wetFloor == 0
    → likelyMurder = true
  - Else
    → likelyMurder = false
- Else
  - If medicineScattered == 1 AND wetFloor == 0
    → likelyMurder = true
  - Else
    → likelyMurder = false

4.1.2. <u>Based on the value of "likelyMurder" variable, print:</u>

- Case status: Likely Murder
- Or, Case status: Likely Accident

Remember: If accident is likey, then your program will stop there. Otherwise, your program will move on to Part 2 and Part 3.

4.2. PART 2− Suspect Evaluation

For EACH of the five suspects, you must compute:

- bool hasMotive;
- bool presentThere;
- bool hasAlibi;
- bool hasFingerprints;
- int suspicionLevel;

Suspicion levels:

- 0 → Innocent
- 1 → Suspicious
- 2 → Prime Suspect
- 3 → Murderer

You must use nested logic to determine the suspicion level.

4.2.1. <u>How to calculate the Suspicion Level? (Suspicion Logic Pattern: Required Structure)</u>

```
if (hasMotive)
{
    if (presentThere)
    {
        if (!hasAlibi)
        {
            if (hasFingerprints)
                suspicionLevel = 3;
            else
                suspicionLevel = 2;
        }
        else
            suspicionLevel = 1;
    }
    else
        suspicionLevel = 1;
}
else
    suspicionLevel = 0;
```

You may not compress this logic.

4.3. PART 3: Interview Questions & Evidence Mapping

Each suspect has different questions. All inputs are 1 (Yes) or 0 (No).

4.3.1. Suspect 1: Mr. Brown (Lawyer)

Mr. Brown is Mr. Ed's lawyer. He is fully aware of all of Mr. Ed's assets and is currently preparing the final version of his will.

Input Evidence Variables:

- willChangeBenefit
- arguedToday
- seenNearHallway
- confirmedPhoneCall
- printsOnDoorHandle

Ask proper questions to the suspect to collect values for the input evidence variables.

i. Would Mr. Brown benefit from a will change?
ii. Did Mr. Brown argue with Mr. Ed today?
iii. Was Mr. Brown seen near the hallway/bathroom area?
iv. Was Mr. Brown on a confirmed phone call at that time (verified alibi)?
v. Were Mr. Brown's fingerprints found on the bathroom door handle?

Evidence rules:

- hasMotive → true if willChangeBenefit OR arguedToday
- presentThere → true if seenNearHallway
- hasAlibi → true if confirmedPhoneCall
- hasFingerprints → true if printsOnDoorHandle

4.3.2. Suspect 2: Parker (Violin Teacher)

Parker, age 37, is Mr. Ed's violin teacher. Mr. Ed always wanted to learn the violin but never had the opportunity during his younger years. After his retirement, he decided to finally fulfill that long-held wish. But sometimes Mr. Ed forgets to pay his fees.

Input Evidence Variables:

- unpaidLessonFees
- heardViolinOutside
- seenLeavingBackyardGate
- printsOnMirror

Ask proper questions to the suspect to collect values for the input evidence variables.

i. Did Mr. Ed owe Parker unpaid lesson fees?
ii. Was violin music heard outside at the time (possible alibi)?
iii. Was Parker seen leaving the backyard gate around that time?
iv. Were Parker's fingerprints found on the bathroom mirror?

Evidence rules:

- hasMotive is true → if unpaidLessonFees is true
- has Alibi is true → if heardViolinOutside is true
- presentThere is true → if seenLeavingBackyardGate is true *and* heardViolinOutside is false
- hasFingerprints is true → printsOnMirror is true

4.3.3. Mark (College Buddy)

Mark is Mr. Ed's college buddy and is also around 74 years old. Although they have been close friends for decades, Mark has often felt overshadowed by Mr. Ed's success. He frequently asks Mr. Ed for financial help.

Input Evidence Variables:

- loanRefused
- drankTooMuch
- seenNearBathroom
- nurseSawWithGroup
- printsOnCabinet

Ask proper questions to the suspect to collect values for the input evidence variables.

i. Did Mr. Ed refuse Mark a loan today?
ii. Did Mark drink too much at the party?
iii. Was Mark seen near the bathroom area?
iv. Did the nurse confirm Mark stayed with the group (verified alibi)?
v. Were Mark's fingerprints found on the bathroom cabinet?

Evidence rules:

- hasMotive → if loanRefused OR drankTooMuch
- presentThere → if seenNearBathroom

- hasAlibi → if nurseSawWithGroup
- hasFingerprints → if printsOnCabinet

Bonus twist:

If drankTooMuch == 1 AND wetFloor == 1 AND heavyObjectMissing == 0 → reduce suspicionLevel by 1 (but not below 0)

4.3.4. Dayan (Chess Partner)

Dayan is a brilliant 22-year-old who lives in Mr. Ed's neighborhood and often visits to play chess with him. Mr. Ed is very fond of Dayan, but Dayan is highly competitive and reacts badly to losing, something that almost never happens.

Input Evidence Variables:

- lostChessMatch
- competitiveAngry
- seenNearBathroom
- wasStreamingOnline
- printsOnFaucet

Ask proper questions to the suspect to collect values for the input evidence variables.

i. Did Dayan lose a chess match to Mr. Ed today?
ii. Was Dayan visibly angry/competitive about losing?
iii. Was Dayan seen near the bathroom area?
iv. Was Dayan confirmed to be streaming online at the time (verified alibi)?
v. Were Dayan's fingerprints found on the bathroom faucet?

Evidence rules:

- hasMotive → true only if lostChessMatch *and* competitiveAngry.
- presentThere → true if seenNearBathroom.
- hasAlibi → true if wasStreamingOnline.
- hasFingerprints → true if printsOnFaucet.

4.3.5. Ms. Elena (Pen Friend)

Ms. Elena is Mr. Ed's long-time pen friend. They had never met in person before, but she arrived unexpectedly to surprise him on his 75th birthday.

Input Evidence Variables:

- secretHistory
- askedPrivateTalk
- stayedWithGroup
- printsOnMedicineBox

Ask proper questions to the suspect to collect values for the input evidence variables.

i. Did Mr. Ed seem shocked/uncomfortable after seeing Ms. Elena (secret history)?
ii. Did Ms. Elena ask Mr. Ed to talk privately?
iii. Did Ms. Elena stay with the group the whole time?
iv. Were Ms. Elena's fingerprints found on a medicine box in the bathroom?

Evidence rules:

- hasMotive → true if Ms. Elena has a secretHistory *or* if she askedPrivateTalk
- hasAlibi → true if she stayedWithGroup
- presentThere → true if she did *not* stayedWithGroup (meaning she could have been near the bathroom).
- hasFingerprints → true if printsOnMedicineBox

4.4. Final Decision Logic

Initialize:

```
int bestLevel = -1;

string bestName = "";
```

After computing suspicionLevel (section 4.2.) for each suspect, check:

```
If suspicionLevel > bestLevel
→ update bestLevel and bestName
```

5. Final Output Rules

```
If likelyMurder == false
 → Print:
    Final Conclusion: No clear suspect (likely accidental case).

Else:

        If bestLevel == 3
        → Print:
            Murderer: <bestName>

        Else if bestLevel == 2
        → Print:
            Prime Suspect: <bestName>

        Else
        → Print:
           No clear suspect.
```

6. Submission Requirements

- Proper indentation
- Clear variable names (Use the same names as used in the assignment document)
- Logical structure
- No giant one-line conditions
- Must include 3 commented test cases:
    - Murder case with a level 3 suspect
    - Murder case with only level 2 suspect
    - Likely accident case

7. Grading Rubric

Scene analysis logic – 20%
Evidence mapping – 25%
Nested suspicion logic – 25%
Final decision logic – 15%
Code readability & structure – 15%

8. Sample outputs

8.1. Murder case with a level 3 suspect

```
C:\Users\royso\source\repos\CrimeSceneInvestigation\x64\Debug\CrimeScen
Is there a wet floor? (1 for Yes, 0 for No): 1
Is there a broken vase? (1 for Yes, 0 for No): 0
Is a heavy object missing? (1 for Yes, 0 for No): 1
Are medicines scattered around? (1 for Yes, 0 for No): 0

Case Status: Likely Murder
```

```
============================
Suspect 1: Mr. Brown (Lawyer)
============================
Would Mr. Brown benefit from a will change? (1/0): 1
Did Mr. Brown argue with Mr. Ed today? (1/0): 0
Was Mr. Brown seen near the hallway/bathroom area? (1/0): 0
Was Mr. Brown on a confirmed phone call at that time (verified alibi)? (1/0): 1
Were Mr. Brown's fingerprints found on the bathroom door handle? (1/0): 1

============================
Suspect 2: Parker (Violin Teacher)
============================
Did Mr. Ed owe Parker unpaid lesson fees? (1/0): 0
Was violin music heard outside at the time (possible alibi)? (1/0): 0
Was Parker seen leaving the backyard gate around that time? (1/0): 0
Were Parker's fingerprints found on the bathroom mirror? (1/0): 0

============================
Suspect 3: Mark (College Buddy)
============================
Did Mr. Ed refuse Mark a loan today? (1/0): 0
Did Mark drink too much at the party? (1/0): 0
Was Mark seen near the bathroom area? (1/0): 0
Did the nurse confirm Mark stayed with the group (verified alibi)? (1/0): 1
Were Mark's fingerprints found on the bathroom cabinet? (1/0): 1

============================
Suspect 4: Dayan (Chess Partner)
============================
Did Dayan lose a chess match to Mr. Ed today? (1/0): 1
Was Dayan visibly angry/competitive about losing? (1/0): 1
Was Dayan seen near the bathroom area? (1/0): 1
Was Dayan confirmed to be streaming online at the time (verified alibi)? (1/0): 0
Were Dayan's fingerprints found on the bathroom faucet? (1/0): 1
```

```
============================
Suspect 5: Ms. Elena (Pen Friend)
============================
Did Mr. Ed seem shocked/uncomfortable after seeing Ms. Elena (secret history)? (1/0): 0
Did Ms. Elena ask Mr. Ed to talk privately? (1/0): 0
Did Ms. Elena stay with the group the whole time? (1/0): 0
Were Ms. Elena's fingerprints found on a medicine box in the bathroom? (1/0): 0

============================
Final Investigation Result
============================
Murderer: Dayan
```

8.2. Murder case with a level 3 suspect

```
C:\Users\royso\source\repos\CrimeSceneInvestigation\x64\Debug\CrimeScer
Is there a wet floor? (1 for Yes, 0 for No): 0
Is there a broken vase? (1 for Yes, 0 for No): 1
Is a heavy object missing? (1 for Yes, 0 for No): 1
Are medicines scattered around? (1 for Yes, 0 for No): 0

Case Status: Likely Murder
```

```
============================
Suspect 1: Mr. Brown (Lawyer)
============================
Would Mr. Brown benefit from a will change? (1/0): 0
Did Mr. Brown argue with Mr. Ed today? (1/0): 1
Was Mr. Brown seen near the hallway/bathroom area? (1/0): 0
Was Mr. Brown on a confirmed phone call at that time (verified alibi)? (1/0): 1
Were Mr. Brown's fingerprints found on the bathroom door handle? (1/0): 0

============================
Suspect 2: Parker (Violin Teacher)
============================
Did Mr. Ed owe Parker unpaid lesson fees? (1/0): 1
Was violin music heard outside at the time (possible alibi)? (1/0): 0
Was Parker seen leaving the backyard gate around that time? (1/0): 1
Were Parker's fingerprints found on the bathroom mirror? (1/0): 0

============================
Suspect 3: Mark (College Buddy)
============================
Did Mr. Ed refuse Mark a loan today? (1/0): 0
Did Mark drink too much at the party? (1/0): 0
Was Mark seen near the bathroom area? (1/0): 0
Did the nurse confirm Mark stayed with the group (verified alibi)? (1/0): 0
Were Mark's fingerprints found on the bathroom cabinet? (1/0): 0

============================
Suspect 4: Dayan (Chess Partner)
============================
Did Dayan lose a chess match to Mr. Ed today? (1/0): 0
Was Dayan visibly angry/competitive about losing? (1/0): 0
Was Dayan seen near the bathroom area? (1/0): 0
Was Dayan confirmed to be streaming online at the time (verified alibi)? (1/0): 0
Were Dayan's fingerprints found on the bathroom faucet? (1/0): 0
```

```
============================
Suspect 5: Ms. Elena (Pen Friend)
============================
Did Mr. Ed seem shocked/uncomfortable after seeing Ms. Elena (secret history)? (1/0): 0
Did Ms. Elena ask Mr. Ed to talk privately? (1/0): 0
Did Ms. Elena stay with the group the whole time? (1/0): 0
Were Ms. Elena's fingerprints found on a medicine box in the bathroom? (1/0): 0

============================
Final Investigation Result
============================
Prime Suspect: Parker
```

8.3. Likely Accident Case

Microsoft Visual Studio Debug Console

```
Is there a wet floor? (1 for Yes, 0 for No): 1
Is there a broken vase? (1 for Yes, 0 for No): 0
Is a heavy object missing? (1 for Yes, 0 for No): 0
Are medicines scattered around? (1 for Yes, 0 for No): 1

Case Status: Likely Accident
Program will stop here (accident is likely).
```